YEAR 12 MATHEMATICS EXTENSION 1

Chapter 10 Related Rates and Projectile Motion

Lesson and Homework Pack

Revision	Motion in one dimension	Advanced · 9A–9C
Lesson 9E	Related rates	Past papers: 20 marks · 30 min
Lesson 10A	Projectile motion: the time equations	Past papers: 20 marks · 30 min
Lesson 10B	Projectile motion: the equation of path	Past papers: 18 marks · 25 min
Review	Chapter 10 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Vectors are written with a tilde, as in $\underset{\sim}{v} = \dot{x}\,\underset{\sim}{i} + \dot{y}\,\underset{\sim}{j}$. Take $g = 10$ m/s² unless a question says otherwise.

Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

Motion in one dimension

A short review of Advanced Sections 9A–9C: displacement, velocity and acceleration as derivatives, and integrating with respect to time. These ideas are used in both directions, horizontal and vertical, throughout projectile motion.

SUMMARY · keep this page open while you work

DISPLACEMENT, VELOCITY AND ACCELERATION (9A, 9B)

- Motion in one dimension is described by the **displacement** x on a number line as a function of the time t after time zero. Negative values of time are excluded unless stated.
- **Average velocity** $= \dfrac{\text{change in displacement}}{\text{change in time}} = \dfrac{x_2 - x_1}{t_2 - t_1}$, the gradient of the chord on the x–t graph. **Average speed** $= \dfrac{\text{distance travelled}}{\text{time taken}}$.
- Distance travelled is not the same as displacement: split the journey at every time the particle changes direction.

$$v = \frac{dx}{dt} = \dot{x} \qquad a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = \ddot{x} \qquad \text{speed} = |v|$$

A particle is **stationary** when $v = 0$. Displacement and velocity are vector quantities (they have a sign); distance and speed are scalars and are never negative.

INTEGRATING WITH RESPECT TO TIME (9C)

- Given the acceleration, integrate to find the velocity. Given the velocity, integrate to find the displacement.
- **Never omit the constants of integration.** Use an initial (boundary) condition to evaluate each one.
- A body falling freely accelerates downwards at $g \doteq 9.8\ \text{m/s}^2$ (or $10\ \text{m/s}^2$ in round figures), provided air resistance is ignored. If upwards is positive, start with $\ddot{x} = -g$ and integrate.

THE CHAIN RULE AND RESOLVING (YEAR 11)

$\dfrac{dy}{dt} = \dfrac{dy}{dx} \times \dfrac{dx}{dt}$ and $\dfrac{dx}{dy} = 1 \Big/ \dfrac{dy}{dx}$. In a right-angled triangle with hypotenuse v and angle θ, the adjacent side is $v\cos\theta$ and the opposite side is $v\sin\theta$.

WARM-UP · answers are in the answers booklet

1 A particle moves so that its displacement in metres from the origin at time t seconds is $x = t^2 - 6t + 5$.

a Find the velocity v and the acceleration a as functions of t.

b When is the particle stationary, and where is it then?

c Find the distance travelled in the first 4 seconds, and the average velocity and average speed over that time.

2 A particle's acceleration is $\ddot{x} = 6t - 4$ m/s². Initially it is at the origin with velocity 3 m/s. Find $\dot{x}$ and x as functions of t, and show that the particle is never stationary.

3 A ball is thrown vertically upwards from the ground with speed 20 m/s. Take upwards as positive and $g = 10\ \text{m/s}^2$. Starting from $\ddot{x} = -10$, find $\dot{x}$ and x, the greatest height, and the time and speed when the ball returns to the ground.

4 Find the exact values of $40\cos 30°$ and $40\sin 30°$. Then find, to the nearest degree, the acute angle θ with $\tan\theta = \frac{3}{4}$, and evaluate $\sqrt{6^2 + 8^2}$.

5 If $A = \pi r^2$ and $\dfrac{dr}{dt} = 3$, find $\dfrac{dA}{dt}$ when $r = 4$.

LESSON 9E

Related rates

Applying the chain rule to rates: expressing one quantity as a function of another, differentiating with respect to time, using geometry to eliminate a variable, rates of change of angles, and chains with more than two links. This reviews Year 11 Section 16A.

CONCEPT 1 · the chain rule applied to rates

RELATED RATES

When two quantities both change with time, and one is a function of the other, their rates of change are related by the chain rule. For example, if a bubble's volume V is a function of its radius r, then

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}.$$

- Express one quantity as a function of the other quantity.
- Then differentiate with respect to time t using the chain rule.
- Substitute the given values **after** differentiating, never before.
- Be very careful with units, here and anywhere calculus is applied. A rate has units such as mm³/s or m/s.

If the rate you are given is for the "wrong" variable, use $\frac{dr}{dV} = 1 \Big/ \frac{dV}{dr}$, or simply solve the chain-rule equation for the unknown rate.

VOLUME AND SURFACE AREA OF SOLIDS (YEAR 11)

	Sphere	Cylinder	Cone	Pyramid
Volume	$V = \frac{4}{3}\pi r^3$	$V = \pi r^2 h$	$V = \frac{1}{3}\pi r^2 h$	$V = \frac{1}{3} \times \text{base} \times \text{height}$
Surface area	$A = 4\pi r^2$	$A = 2\pi r^2 + 2\pi r h$	$A = \pi r^2 + \pi r \ell$	$A =$ sum of faces

In the formula for the surface area of a cone, ℓ is the slant height.

TUTOR DEMO **A growing sphere**

A spherical balloon is being inflated, so that $V = \frac{4}{3}\pi r^3$.

a Differentiate with respect to time using the chain rule.

b Find the rate of increase of the volume when the radius is 5 mm, if the radius is increasing at 2 mm/s.

c Find the rate of increase of the radius when the volume is 36π mm³, if the volume is increasing at 72π mm³/s.

YOUR TURN **Same method**

The sides of a square of side length x metres are increasing at 0.2 m/s.

a Find the rate of increase of the area when $x = 4$.

b What is the side length when the area is increasing at 2 m²/s?

WORKED EXAMPLE 1 **Keep for homework**

A hemispherical bubble on the water surface is growing in size.
a Write down the formula for the volume V mm³ in terms of the radius r mm, then differentiate with respect to time using the chain rule.
b Find the rate at which the volume is increasing when the radius is 6 mm, if the radius is increasing at 3 mm/s.
c Find the rate at which the radius is increasing when the volume is 18 mm³, if the volume is increasing at 90 mm³/s.

SOLUTION

a The volume is $V = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2\pi}{3} r^3$.

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = 2\pi r^2 \frac{dr}{dt}$$

b Substituting $r = 6$ and $\frac{dr}{dt} = 3$:

$$\frac{dV}{dt} = 2\pi \times 36 \times 3 = 216\pi \text{ mm}^3/\text{s}.$$

c When $V = 18$: $18 = \frac{2\pi}{3} r^3$, so $r^3 = 27\pi^{-1}$ and $r = 3\pi^{-\frac{1}{3}}$.

Substituting $r = 3\pi^{-\frac{1}{3}}$ and $\frac{dV}{dt} = 90$:

$$90 = 2\pi \times 9\pi^{-\frac{2}{3}} \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = 5\pi^{-\frac{1}{3}} \text{ mm/s} \doteq 3.4 \text{ mm/s}.$$

CONCEPT 2 · using the geometry to eliminate a variable

ONE VARIABLE TOO MANY

Often the formula involves two changing lengths, such as the radius r and depth h of water in a cone, $V = \frac{1}{3}\pi r^2 h$. Before differentiating, use the geometry of the situation (similar triangles, a given ratio, or Pythagoras' theorem) to write the quantity as a function of **one** variable only.

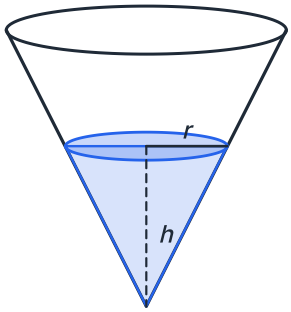
- Draw a diagram and name every quantity that changes.
- Write down the rate you are given and the rate you want, with their signs. A quantity that is decreasing has a **negative** rate.
- Find the relation between the two quantities in one variable, then differentiate with respect to t by the chain rule.

TUTOR DEMO **Filling a cone**

Water is poured at 12 cm³/s into a conical tank with its vertex down. When the depth of water is h cm, the radius of the water surface is $r = \frac{1}{2}h$ cm.

a Show that the volume of water is $V = \frac{\pi}{12}h^3$.

b Find the rate at which the depth is increasing when the depth is 4 cm.

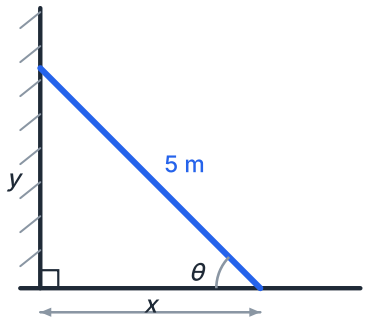


YOUR TURN **Same method**

Water flows at 9π cm³/s into an inverted cone in which the radius of the water surface always equals the depth, $r = h$. Find the rate at which the depth is increasing when the depth is 3 cm.

WORKED EXAMPLE 2 **Keep for homework**

A 5-metre ladder leans against a vertical wall. Its foot slides away from the wall at 0.3 m/s. Find the rate at which the top of the ladder is moving down the wall when the foot is 3 m from the wall.



SOLUTION

Let the foot be x m from the wall and the top be y m above the ground. We are given $\frac{dx}{dt} = 0.3$, and we want $\frac{dy}{dt}$.

By Pythagoras,

$y = \sqrt{25 - x^2} = (25 - x^2)^{\frac{1}{2}}$, so

$$\frac{dy}{dx} = \tfrac{1}{2}(25 - x^2)^{-\frac{1}{2}} \times (-2x)$$
$$= \frac{-x}{\sqrt{25 - x^2}}.$$

By the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$
$$= \frac{-x}{\sqrt{25 - x^2}} \times 0.3.$$

When $x = 3$, $\sqrt{25 - 9} = 4$, so

$$\frac{dy}{dt} = -\frac{3}{4} \times 0.3 = -0.225 \text{ m/s}.$$

The top of the ladder is sliding down the wall at 0.225 m/s. The negative sign shows that y is decreasing.

CONCEPT 3 · rates of change of angles

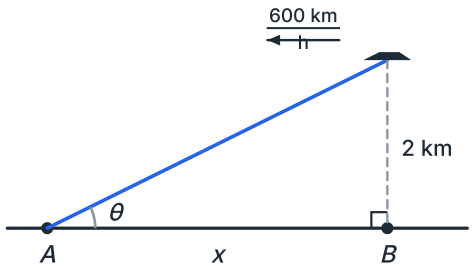
ANGLES THAT CHANGE WITH TIME

- Use trigonometry to express a length in terms of the angle, for example $x = \frac{h}{\tan\theta}$, then differentiate with respect to θ.
- Calculus with trigonometric functions always uses **radians**, so $\frac{d\theta}{dt}$ comes out in radians per unit of time.
- To convert, multiply by $\frac{180}{\pi}$ to get degrees, and divide by 60 or 3600 to change minutes or hours into seconds.

$$\frac{d}{d\theta}\left(\frac{1}{\tan\theta}\right) = \frac{d}{d\theta}\left(\frac{\cos\theta}{\sin\theta}\right) = -\frac{1}{\sin^2\theta} \qquad \frac{d}{d\theta}(\tan\theta) = \sec^2\theta = \frac{1}{\cos^2\theta}$$

TUTOR DEMO **Watching a plane**

A plane flies at 600 km/h at a constant altitude of 2 km, directly towards an observer at A. Let θ be its angle of elevation from A, and let x km be the distance from A to the point B directly below the plane.



a By writing $x = \dfrac{2}{\tan\theta}$, show that $\dfrac{dx}{d\theta} = -\dfrac{2}{\sin^2\theta}$.

b Find the rate at which θ is increasing when $\theta = \dfrac{\pi}{6}$, in radians per hour and in degrees per second.

YOUR TURN **Same method**

A boat is observed from the top of an 80-metre-high cliff. The boat travels directly towards the cliff at 4 m/s. Find the rate at which the angle of depression is changing when it is $30°$, in radians per second and in degrees per second (to two decimal places).

WORKED EXAMPLE 3 **Keep for homework**

A balloon rises vertically at 5 m/s from a point B on level ground. An observer stands at A, 120 m from B. Find the rate at which the angle of elevation θ of the balloon from A is increasing when the balloon is 50 m high.

SOLUTION

Let the height be h m. Then $h = 120\tan\theta$, so $\dfrac{dh}{d\theta} = 120\sec^2\theta$ and

$$\frac{d\theta}{dh} = \frac{1}{120\sec^2\theta} = \frac{\cos^2\theta}{120}.$$

By the chain rule,

$$\frac{d\theta}{dt} = \frac{d\theta}{dh} \times \frac{dh}{dt} = \frac{\cos^2\theta}{24}.$$

When $h = 50$, the line of sight has length $\sqrt{120^2 + 50^2} = 130$, so $\cos\theta = \dfrac{12}{13}$ and

$$\frac{d\theta}{dt} = \frac{144}{169} \times \frac{1}{24} = \frac{6}{169}.$$

The angle is increasing at $\dfrac{6}{169} \doteq 0.0355$ radians per second.

In degrees, $\dfrac{6}{169} \times \dfrac{180}{\pi} \doteq 2.03°$ per second.

Chains can have more than two links. If an area A depends on an angle θ, which depends on a length x, which changes with time, then $\dfrac{dA}{dt} = \dfrac{dA}{d\theta} \times \dfrac{d\theta}{dx} \times \dfrac{dx}{dt}$ (see Question 12).

TEXTBOOK HOMEWORK · Exercise 9E

FOUNDATION

1 The sides of a square of side length x metres are increasing at a rate of 0.1 m/s.

a Show that the rate of increase of the area is given by $\frac{dA}{dt} = 0.2x$ m²/s.

b At what rate is the area of the square increasing when its sides are 5 metres long?

c What is the side length when the area is increasing at 1.4 m²/s?

d What is the area when the area is increasing at 0.6 m²/s?

2 The diagonal of a square is decreasing at a rate of $\frac{1}{2}$ m/s.

a Find the area A of a square with a diagonal of length ℓ.

b Hence show that the rate of change of area is $\frac{dA}{dt} = -\frac{1}{2}\ell$ m²/s.

c Find the rate at which the area is decreasing when: (i) the diagonal is 10 metres, (ii) the area is 18 m².

d What is the length of the diagonal when the area is decreasing at 17 m²/s?

3 The radius r of a sphere is increasing at a rate of 0.3 m/s. In both parts, approximate π using a calculator and give your answer correct to three significant figures.

a Show that the sphere's rate of change of volume is $\dfrac{dV}{dt} = 1.2\pi r^2$, and find the rate of increase of its volume when the radius is 2 metres.

b Show that the sphere's rate of change of surface area is $\dfrac{dS}{dt} = 2.4\pi r$, and find the rate of increase of its surface area when the radius is 4 metres.

4 Jules is blowing up a spherical balloon at a constant rate of 200 cm^3/s.

a Show that $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$.

b Hence find the rate at which the radius is growing when the radius is 15 cm.

c Find the radius and volume when the radius is growing at 0.5 cm/s.

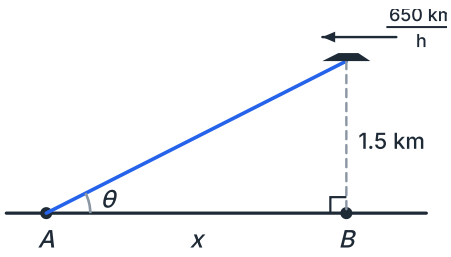
DEVELOPMENT

5 An artist pours plaster of Paris into a conical mould at a rate of 5 cm^3/s. At time t, let the top of the plaster have radius r, and let the depth be h, where $h = 2r$.

a Find the volume of plaster, and show that it is the same as that of a hemisphere with the same radius.

b Find the rate at which the radius is increasing when the depth of the plaster is 10 cm.

6 An observer at A in the diagram is watching a plane fly overhead, and he tilts his head so that he is always looking directly at the plane. The aircraft is flying at 650 km/h at an altitude of 1.5 km. Let θ be the angle of elevation of the plane from the observer, and suppose that the distance from A to B, directly below the aircraft, is x km.

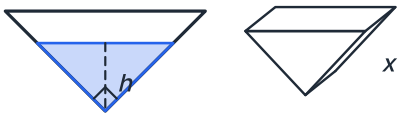


a By writing $x = \dfrac{3}{2\tan\theta}$, show that $\dfrac{dx}{d\theta} = -\dfrac{3}{2\sin^2\theta}$.

b Hence find the rate at which the observer's head is tilting when the angle of inclination to the plane is $\frac{\pi}{3}$. Convert your answer from radians per hour to degrees per second, correct to the nearest degree.

7 A boat is observed from the top of a 100-metre-high cliff. The boat is travelling towards the cliff at a speed of 50 m/min. How fast is the angle of depression changing when the angle of depression is $15°$? Convert your answer from radians per minute to degrees per minute, correct to the nearest degree.

8 The water trough in the diagram is in the shape of an isosceles right triangular prism, x metres long. It is found that during summer in a drought, the amount of water lost to evaporation each day in cm³ is equal to 10% of the surface area in cm².



cross–section

a Find a formula for the volume V cm³ of water in the trough when the depth is h cm.

b Show that the rate at which the depth of the water is changing is constant.

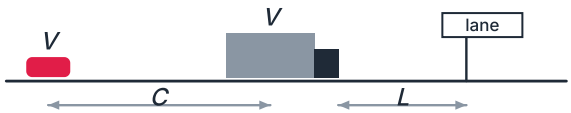
9 The volume of a sphere is increasing at a rate numerically equal to its surface area at that instant. Show that $\frac{dr}{dt} = 1$.

10 A point moves anticlockwise around the circle $x^2 + y^2 = 1$ at a uniform speed of 2 m/s.

a Find an expression for the rate of change of its x-coordinate in terms of x, when the point is above the x-axis. (The units on the axes are metres.)

b Use your answer to part a to find the rate of change of the x-coordinate as it crosses the y-axis at $P(0, 1)$. Why should this answer have been obvious without this formula?

11 A car is travelling C metres behind a truck, both travelling at a constant speed of V m/s. The road widens L metres ahead of the truck and there is an overtaking lane. The car accelerates at a uniform rate so that it is exactly alongside the truck at the beginning of the overtaking lane.



a What is the acceleration of the car?

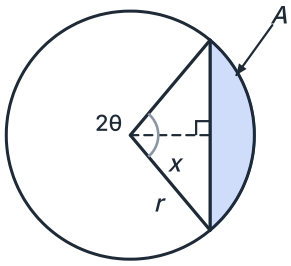
b Show that the speed of the car as it passes the truck is $V\left(1+\dfrac{2C}{L}\right)$.

c The objective of the driver of the car is to spend as little time alongside the truck as possible. What strategies could the driver employ?

d The speed limit is 100 km/h, and the truck is travelling at 90 km/h and is 50 metres ahead of the car. How far before the overtaking lane should the car begin to accelerate if applying the objective in part c?

ENRICHMENT

12 The diagram shows a chord distant x from the centre of a circle. The radius of the circle is r, and the chord subtends an angle 2θ at the centre.



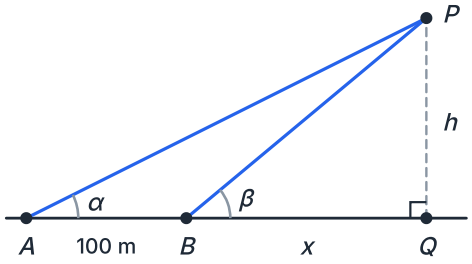
a Show that the area of the segment cut off by this chord is $A = r^2(\theta - \sin\theta\cos\theta)$.

b Explain why $\dfrac{dA}{dt} = \dfrac{dA}{d\theta} \times \dfrac{d\theta}{dx} \times \dfrac{dx}{dt}$.

c Show that $\dfrac{d\theta}{dx} = -\dfrac{1}{\sqrt{r^2 - x^2}}$.

d Given that $r = 2$, find the rate of increase in the area if $\dfrac{dx}{dt} = -\sqrt{3}$ when $x = 1$.

13 The diagram shows two radars at A and B 100 metres apart. An aircraft at P is approaching and the radars are tracking it, hence the angles α and β are changing with time.



a Show that $x \tan \beta = (x + 100) \tan \alpha$.

b Keeping in mind that x, α and β are all functions of time, use the chain and product rules to show that $\dfrac{dx}{dt} = \dfrac{\dot{\alpha}(x + 100) \sec^2 \alpha - \dot{\beta} x \sec^2 \beta}{\tan \beta - \tan \alpha}$.

c Use part a to find the value of x and the height of the plane when $\alpha = \frac{\pi}{6}$ and $\beta = \frac{\pi}{4}$.

d At the angles given in part c, it is found that $\dfrac{d\alpha}{dt} = \frac{5}{36}\left(\sqrt{3} - 1\right)$ radians per second and $\dfrac{d\beta}{dt} = \frac{5}{18}\left(\sqrt{3} - 1\right)$ radians per second. Find the speed of the plane.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q2** **1 mark**

A spherical balloon is being inflated at a constant rate of 200π cm^3 s^{-1}. At what rate is the radius of the balloon increasing when the radius is 10 cm?

(A) 0.25 cm s^{-1} (B) 0.5 cm s^{-1} (C) 1.0 cm s^{-1} (D) 2.0 cm s^{-1}

Hint: $V = \frac{4}{3}\pi r^3$. Use $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$.

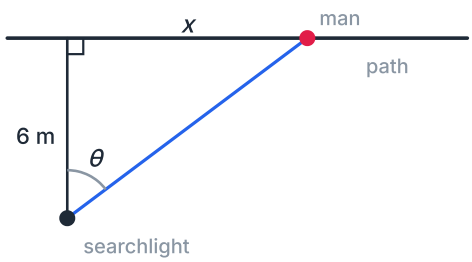
PP 2 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d)** **2 marks**

Tea is spilling onto the floor, forming an expanding circle.
The radius of this circle increases at a rate of 2 cm/s.
At what rate is the surface area of the tea increasing when the radius is 8 cm?

Hint: the area of the circle is $A = \pi r^2$. Find $\frac{dA}{dr}$ first.

PP 3 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(e)** 3 marks

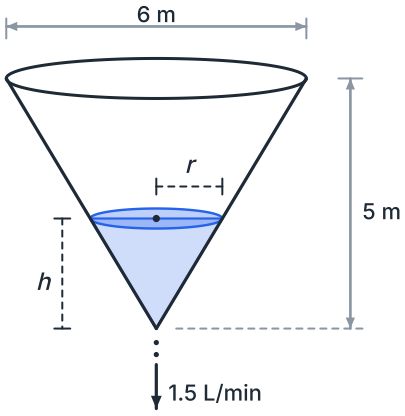
A man walks along a straight path at a speed of 1.2 m/s. A searchlight is located on the ground 6 m from the path and is kept focused on the man.
At what rate is the searchlight rotating when the man is 8 m from the point on the path closest to the searchlight?



Hint: from the diagram, $\tan\theta = \frac{x}{6}$. Differentiate both sides with respect to t.

PP 4 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q14(c)** 3 marks

A tank in the shape of a cone is filled with water, which is leaking out from the bottom at a rate of 1.5 litres per minute.
The volume of water in the tank is given by $V = \frac{1}{3}\pi r^2 h$, where h is the height in metres and r is the radius in metres.
Find the rate at which the water level is falling when the height of the water is 1 metre.
[Note: 1 m^3 = 1000 L]

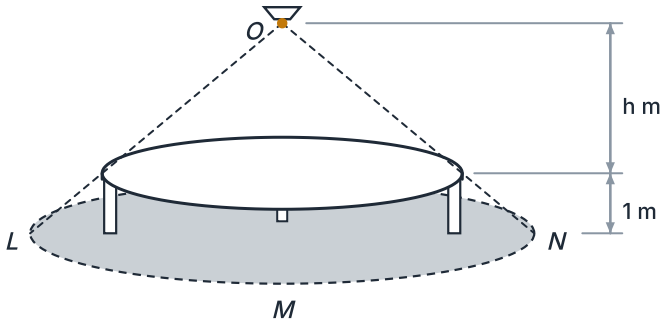


Hint: use similar triangles to write r in terms of h, so that V is a function of h only. Change litres to m^3.

PP 5 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q13(d)** **5 marks**

A small lamp O is placed h m above the centre of a round table of radius 3 m and height of 1 m. The lamp casts a circular shadow LMN of the table on the ground as shown in the diagram below.

At any given time, let h be the height, in metres, of the lamp above the table and let A be the area of the shadow in m^2.



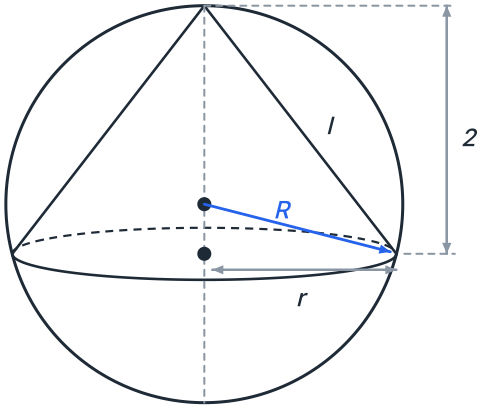
(i) Show that $A = 9\pi\left(1 + \dfrac{1}{h}\right)^2$. 2

(ii) If the lamp is lowered vertically at a constant rate of $\dfrac{3}{100}$ m/s, find the exact rate of change of A with respect to time when the lamp is 3 m above the table. 3

Hint: in (i), take a vertical cross-section through O and the centre of the table, and use similar triangles. In (ii), h is decreasing, so $\dfrac{dh}{dt}$ is negative.

PP 6 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q12(b)** 6 marks

The diagram shows a right circular cone of height 2 units and radius r and slant height l inscribed in a sphere of radius R.



(i) Show that $A = 4\pi\sqrt{R^2 - R}$, where A is the curved surface area of the cone. 2
(Curved surface area of a right circular cone of base radius r and slant height l is πrl)

(ii) If the volume of the sphere is increasing at the rate of 8 units^3/sec, find the exact rate of change of A at the instant when $R = 2$ units. 4

(Volume of sphere is $\frac{4}{3}\pi R^3$)

Hint: in (i), the centre of the sphere is on the axis of the cone, R below the vertex. Use Pythagoras to find r^2 and then l^2 in terms of R. In (ii), find $\frac{dR}{dt}$ from the volume first.

LESSON 10A

Projectile motion: the time equations

A projectile is thrown or fired into the air and then moves under gravity alone. We treat it as a point, ignore air resistance, and take $g \doteq 9.8$ m/s² (or 10 m/s²). This lesson covers the two ways to describe velocity, resolving velocity, the fundamental equations $\ddot{x} = 0$ and $\ddot{y} = -g$, and problems where the speed or angle is unknown.

CONCEPT 1 · two ways to specify velocity

POSITION AND VELOCITY IN TWO DIMENSIONS

- Place the origin at the point of projection. Measure horizontal distance x and vertical distance y from it. Treat time t as a **parameter**: at each time t the projectile is at some point $P(x, y)$ on its path.
- Take $\underset{\sim}{i}$ and $\underset{\sim}{j}$ as unit vectors in the positive x and y directions. The point $P(30, 15)$ then has position vector $\overrightarrow{OP} = 30\underset{\sim}{i} + 15\underset{\sim}{j}$.
- Velocity is a vector. Its **speed** is its magnitude.

The velocity at any time t can be given in two ways:

- By its **speed and angle of inclination**. The angle of inclination is the acute angle between the path and the horizontal. It is positive if the object is travelling upwards and negative if it is travelling downwards.
- By its **horizontal and vertical components** $\dot{x}$ and $\dot{y}$, so that $\underset{\sim}{v} = \dot{x}\,\underset{\sim}{i} + \dot{y}\,\underset{\sim}{j}$, where $\dot{x} = \underset{\sim}{v} \cdot \underset{\sim}{i}$ and $\dot{y} = \underset{\sim}{v} \cdot \underset{\sim}{j}$.

RESOLUTION OF VELOCITY

To convert between the two descriptions, draw a diagram to **resolve the velocity into its horizontal and vertical components**, or use the conversion equations

$$\begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \end{cases} \quad \text{and} \quad \begin{cases} v^2 = \dot{x}^2 + \dot{y}^2 \\ \tan\theta = \dfrac{\dot{y}}{\dot{x}} \end{cases}$$

(where ambiguity between θ and $180° - \theta$ may need to be clarified by the diagram).

TUTOR DEMO **Resolving, both ways**

a A projectile moves with speed 20 m/s. Resolve its velocity into components, and write it as a vector, when its angle of inclination is (i) $30°$ (ii) $-45°$.

b Find the speed and the angle of inclination (to the nearest degree) when the velocity is (i) $\underset{\sim}{v} = 6\underset{\sim}{i} + 8\underset{\sim}{j}$ (ii) $\underset{\sim}{v} = 12\underset{\sim}{i} - 5\underset{\sim}{j}$.

YOUR TURN **Same skills**

a Resolve a velocity of 16 m/s at an angle of inclination of (i) $60°$ (ii) $-30°$, giving each answer as a vector.

b Find the speed and angle of inclination when (i) $\underset{\sim}{v} = 5\underset{\sim}{i} + 12\underset{\sim}{j}$ (ii) $\underset{\sim}{v} = 7\underset{\sim}{i} - 7\underset{\sim}{j}$.

WORKED EXAMPLE 1 **Keep for homework**

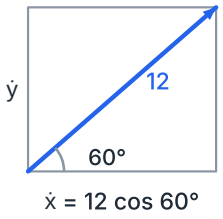
a Resolve the velocity of a projectile moving at 12 m/s, with angle of inclination (i) $60°$ (ii) $-60°$, giving the velocity as a vector.

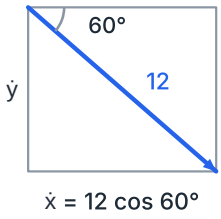
b Find the speed v and angle of inclination θ (to the nearest degree) when (i) $\underset{\sim}{v} = 4\underset{\sim}{i} + 3\underset{\sim}{j}$ m/s (ii) $\underset{\sim}{v} = 5\underset{\sim}{i} - 2\underset{\sim}{j}$ m/s.

SOLUTION

a (i) $\dot{x} = 12\cos 60° = 6$ m/s and $\dot{y} = 12\sin 60° = 6\sqrt{3}$ m/s, so $\underset{\sim}{v} = 6\underset{\sim}{i} + 6\sqrt{3}\,\underset{\sim}{j}$ m/s.

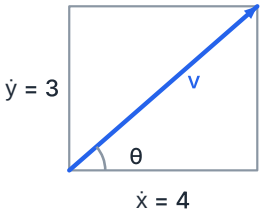
(ii) $\dot{x} = 12\cos 60° = 6$ m/s and $\dot{y} = -12\sin 60° = -6\sqrt{3}$ m/s, so $\underset{\sim}{v} = 6\underset{\sim}{i} - 6\sqrt{3}\,\underset{\sim}{j}$ m/s.

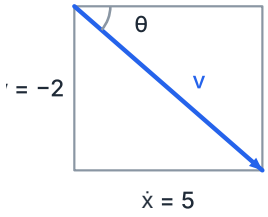




b (i) $v^2 = 4^2 + 3^2$, so $v = 5$ m/s, and $\tan\theta = \frac{3}{4}$, so $\theta \doteq 37°$.

(ii) $v^2 = 5^2 + 2^2$, so $v = \sqrt{29}$ m/s, and $\tan\theta = -\frac{2}{5}$, so $\theta \doteq -22°$.





CONCEPT 2 · the fundamental equations of projectile motion

THE VERTICAL AND HORIZONTAL MOTIONS ARE INDEPENDENT

No force acts on a projectile except the downwards force of gravity. This acceleration is downwards, so it affects the vertical component $\dot{y}$ of the velocity according to $\ddot{y} = -g$. It has no effect on the horizontal component $\dot{x}$, so $\ddot{x} = 0$. In vector form, $\underset{\sim}{a} = -g\underset{\sim}{j}$.

- Unless otherwise indicated, **every question on projectile motion should begin with the two equations $\ddot{x} = 0$ and $\ddot{y} = -g$.**
- The working usually involves four integrations, two for $\ddot{x}$ and two for $\ddot{y}$, with four corresponding substitutions of the initial conditions (resolve the initial velocity first).
- This gives six equations: the two accelerations, the two velocities $\dot{x}$, $\dot{y}$, and the two displacements x, y. Number them, so you can refer to them.

Key ideas: at the greatest height, $\dot{y} = 0$. The projectile lands when y equals the height of the ground (0, or negative if it lands below the origin). The horizontal range is the value of x when it lands. To find the impact velocity, resolve $\dot{x}$ and $\dot{y}$ back into speed and angle.

TUTOR DEMO **The six equations, from a platform**

A ball is thrown with initial speed 30 m/s at an angle of inclination $30°$ from the top of a platform 20 metres above the ground. Take $g = 10$ m/s² and the top of the platform as the origin.

a Find the six equations of motion.

b Find the greatest height above the platform, and when it is reached.

c Find the flight time, the horizontal range, and the impact speed and angle (to the nearest degree).

YOUR TURN **Same method, level ground**

A ball is thrown from ground level with initial speed 20 m/s at an angle of $30°$ to the horizontal. Take $g = 10$ m/s². Find the six equations of motion, the greatest height, the flight time and the horizontal range.

WORKED EXAMPLE 2 **Keep for homework**

A ball is thrown with initial speed 40 m/s and angle of inclination $30°$ from the top of a stand 25 metres above the ground.
a Using the stand as the origin and $g = 10$ m/s², find the six equations of motion.
b Find how high it rises, how long it takes to get there, what its speed is then, and how far it is horizontally from the stand.
c Find the flight time, the horizontal range, and the impact speed and angle.

SOLUTION

Initially $x = y = 0$, and $\dot{x} = 40\cos 30° = 20\sqrt{3}$, $\dot{y} = 40\sin 30° = 20$.

a To begin, $\ddot{x} = 0$. (1)

Integrating, $\dot{x} = C_1$. When $t = 0$, $\dot{x} = 20\sqrt{3}$, so $C_1 = 20\sqrt{3}$ and

$$\dot{x} = 20\sqrt{3}. \quad (2)$$

Integrating, $x = 20\sqrt{3}\,t + C_2$. When $t = 0$, $x = 0$, so $C_2 = 0$ and

$$x = 20\sqrt{3}\,t. \quad (3)$$

b At the top of its flight, the vertical component of velocity is zero, so put $\dot{y} = 0$. From (5), $t = 2$ seconds. From (6), $y = -20 + 40 = 20$ metres above the stand, and from (3), $x = 40\sqrt{3}$ metres from the stand. Because $\dot{y} = 0$ there, the speed is $\dot{x} = 20\sqrt{3}$ m/s.

c It hits the ground when it is 25 metres below the stand, so put $y = -25$. From (6), $-5t^2 + 20t = -25$, so $t^2 - 4t - 5 = 0$, $(t-5)(t+1) = 0$, and $t = 5$ ($t = -1$ is inadmissible). From (3), the horizontal range is $x = 100\sqrt{3}$ metres.

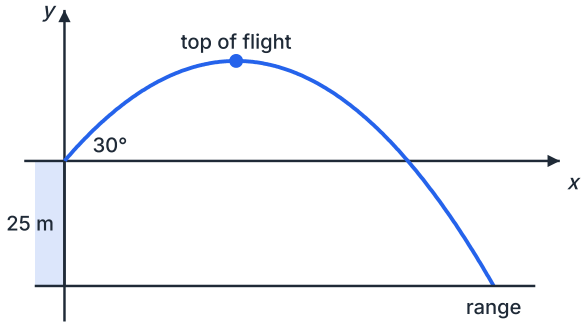
To begin, $\ddot{y} = -10$. (4)

Integrating, $\dot{y} = -10t + C_3$. When $t = 0$, $\dot{y} = 20$, so $C_3 = 20$ and

$$\dot{y} = -10t + 20. \quad (5)$$

Integrating, $y = -5t^2 + 20t + C_4$. When $t = 0$, $y = 0$, so $C_4 = 0$ and

$$y = -5t^2 + 20t. \quad (6)$$



At impact, $\dot{x} = 20\sqrt{3}$ and $\dot{y} = -50 + 20 = -30$, so $v^2 = 1200 + 900$ and the impact speed is $v = 10\sqrt{21}$ m/s. Also $\tan\theta = -\dfrac{30}{20\sqrt{3}}$, so $\theta \doteq -40°54'$: the impact angle is about $40°54'$ below the horizontal.

CONCEPT 3 · using pronumerals for the initial speed or angle

UNKNOWN ANGLE OF PROJECTION

Many problems require the initial speed or angle of inclination to be found so that the projectile behaves in some particular way. Often the muzzle speed of a gun is fixed but the angle α can be altered. There are then usually **two** solutions: a low-flying shot and a "lobbed" shot that goes high into the air.

- Resolve with pronumerals: $\dot{x} = V\cos\alpha$, $\dot{y} = V\sin\alpha$, and derive the six equations as before.
- Substitute the target's x into the x-equation to find t in terms of α, then substitute into the y-equation.
- Use $\dfrac{1}{\cos^2\alpha} = \sec^2\alpha = 1 + \tan^2\alpha$ to obtain a **quadratic in $\tan\alpha$**.

TUTOR DEMO **Two angles to hit a target**

A gun at O fires shells with initial speed 50 m/s, at a variable angle of inclination α. Take $g = 10$ m/s².

a Find the two possible angles (to the nearest minute where necessary) that will hit a target 200 m away horizontally and 40 m above the gun.

b Find the flight time for each angle.

YOUR TURN **Same method**

A cannon fires with initial speed 40 m/s at a variable angle α. Take $g = 10$ m/s². Find the two angles at which it can be fired to hit a target 80 m away horizontally and 40 m above the cannon.

WORKED EXAMPLE 3 **Keep for homework**

A gun at O fires shells with an initial speed of 200 m/s, but a variable angle of inclination α. Take $g = 10$ m/s².

a Find the two possible angles at which the gun can be set so that it will hit a fortress F that is 2 km away on top of a mountain 1000 metres high.

b Show that the inclination of the lower angle to OF is the same as the inclination of the higher angle to the vertical.

c Find the corresponding flight times and the impact speeds and angles.

SOLUTION

Place the origin at the gun. Resolving, $\dot{x} = 200\cos\alpha$ and $\dot{y} = 200\sin\alpha$. Integrating from $\ddot{x} = 0$ and $\ddot{y} = -10$ exactly as before:

$\dot{x} = 200\cos\alpha \quad (2)$ $\qquad$ $x = 200t\cos\alpha \quad (3)$

$\dot{y} = -10t + 200\sin\alpha \quad (5)$ $\qquad$ $y = -5t^2 + 200t\sin\alpha \quad (6)$

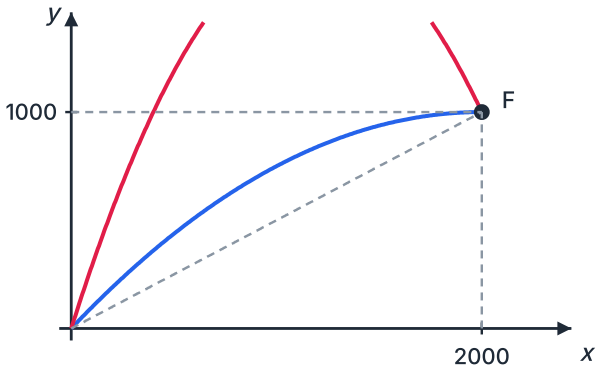
a Because the fortress is 2 km away, $x = 2000$, so from (3), $t = \dfrac{10}{\cos\alpha}$.

Because the mountain is 1000 m high, $y = 1000$, so from (6),

$$-\frac{500}{\cos^2\alpha} + \frac{2000\sin\alpha}{\cos\alpha} - 1000 = 0$$

$$\sec^2\alpha - 4\tan\alpha + 2 = 0.$$

But $\sec^2\alpha = \tan^2\alpha + 1$, so $\tan^2\alpha - 4\tan\alpha + 3 = 0$, $(\tan\alpha - 3)(\tan\alpha - 1) = 0$, giving $\tan\alpha = 1$ or 3, and $\alpha = 45°$ or $\tan^{-1}3 \doteq 71°34'$.



b The line OF is inclined at $\tan^{-1}\frac{1}{2} \doteq 26°34'$ to the horizontal, so the $45°$ shot is inclined at $18°26'$ to OF, and the $71°34'$ shot is inclined at $18°26'$ to the vertical.

c When $\alpha = 45°$: $t = \dfrac{10}{\cos 45°} = 10\sqrt{2}$ seconds. Then from (5), $\dot{y} = -100\sqrt{2} + 200 \times \frac{1}{\sqrt{2}} = 0$, so from (2) the shell hits **horizontally** at $100\sqrt{2}$ m/s.

When $\alpha = \tan^{-1}3$: $\cos\alpha = \dfrac{1}{\sqrt{10}}$ and $\sin\alpha = \dfrac{3}{\sqrt{10}}$, so $t = \dfrac{10}{\cos\alpha} = 10\sqrt{10}$ seconds. From (5), $\dot{y} = -100\sqrt{10} + 60\sqrt{10} = -40\sqrt{10}$, and from (2), $\dot{x} = 20\sqrt{10}$.

Then $v^2 = 16\,000 + 4000 = 20\,000$, so $v = 100\sqrt{2}$ m/s. Also $\tan\theta = \dfrac{\dot{y}}{\dot{x}} = -2$, so $\theta = -\tan^{-1}2 \doteq -63°26'$. This shell hits at $100\sqrt{2}$ m/s, at about $63°26'$ below the horizontal.

TEXTBOOK HOMEWORK · Exercise 10A

Note: in this exercise take $g = 10$ m/s² unless otherwise indicated.

FOUNDATION

1 A particle is projected from the origin with a speed of $30\sqrt{2}$ m/s at an angle of $45°$ to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 30$ and $\dot{y} = -10t + 30$.

b Hence find x and y in terms of t.

c Put $y = 0$ to find when the particle returns to the x-axis.

d Hence find the horizontal distance travelled by the particle.

e Put $\dot{y} = 0$ to find when the particle reaches its greatest height above the x-axis.

f Find the greatest height.

2 A particle is projected from horizontal ground with a speed of 40 m/s at an angle of $30°$ to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 20\sqrt{3}$ and $\dot{y} = -10t + 20$.

b Hence find x and y in terms of t.

c Find: (i) when the particle returns to the ground, (ii) the horizontal distance travelled by the particle, (iii) the greatest height reached above the ground.

3 A particle is projected from the origin with a speed of 20 m/s at an angle of $60°$ to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 10$ and $\dot{y} = -10t + 10\sqrt{3}$.

b Hence find x and y in terms of t.

c Use Pythagoras' theorem to find, correct to one decimal place: (i) the distance of the particle from the origin after one second, (ii) the speed of the particle after one second.

4 A particle is projected from ground level with a speed of 60 m/s at an angle of $\tan^{-1}\frac{4}{3}$ to the horizontal.

a Show that $\dot{x} = 36$ and $\dot{y} = -10t + 48$.

b Find x and y as functions of t.

c Find, correct to one decimal place, the distance of the particle from the point of projection after 3 seconds.

d Show that the velocity of the particle after 3 seconds is $18\sqrt{5}$ m/s at an angle of $\tan^{-1}\frac{1}{2}$ above the horizontal. (Note that velocity is a vector quantity, so both the speed and the direction must be specified.)

5 A particle is projected from the origin. Its initial velocity vector is $8\underset{\sim}{i} + 6\underset{\sim}{j}$.

a Express its velocity at time t in the form $\underset{\sim}{v} = \dot{x}\underset{\sim}{i} + \dot{y}\underset{\sim}{j}$.

b Express its displacement at time t in the form $\underset{\sim}{r} = x\underset{\sim}{i} + y\underset{\sim}{j}$.

c Find: (i) the initial speed of the particle, (ii) the position vector of the particle after 2 seconds, (iii) the position vector of the particle when it reaches its greatest height.

6 A particle is projected from the origin. Its horizontal and vertical components of displacement after t seconds are $x = 40t$ and $y = -5t^2 + 25t$ respectively.

a Find the initial values of $\dot{x}$ and $\dot{y}$.

b Hence show that the initial velocity is $5\sqrt{89}$ at an angle of approximately $32°$ to the horizontal.

DEVELOPMENT

7 A particle is projected from the origin with initial speed V at an angle of α to the horizontal. Two seconds later it passes through the point with position vector $8\underset{\sim}{i} - 12\underset{\sim}{j}$.

a Show that $V\cos\alpha = 4$ and $V\sin\alpha = 4$, and hence write down the initial velocity vector of the particle.

b Find the position vector of the particle after 0.5 seconds.

8 A particle is projected from a point O with initial speed V m/s at an angle of θ to the horizontal. After 2 seconds its horizontal and vertical displacements from O are both 30 m.

a Show that $V\cos\theta = 15$ and $V\sin\theta = 25$.

b Hence show that $V = 5\sqrt{34}$ m/s and $\theta = \arctan\frac{5}{3}$.

9 A stone is thrown from the top of an 11 m high vertical tower standing on level ground. The initial speed is 12 m/s and the initial direction is $30°$ *below* the horizontal.

a Starting with the horizontal and vertical components of acceleration, show that $\dot{x} = 6\sqrt{3}$ and $\dot{y} = -10t - 6$.

b If the origin is at the point of projection, find x and y in terms of t.

c How long will it take for the stone to hit the ground?

d How far from the base of the tower, in metres correct to one decimal place, will the stone hit the ground?

e Show that the stone will hit the ground at an angle of $\tan^{-1}\dfrac{8\sqrt{3}}{9}$ below the horizontal.

10 A ball is tossed with initial speed 6 m/s at an angle of $45°$ to the horizontal. It hits the ground at a point that is 2 m below the point of projection. Find, correct to two decimal places: **a** the time of flight of the ball, **b** the horizontal distance travelled by the ball.

11 A projectile was fired from level ground and just cleared a wall that is 10 m high and 20 m from the point of projection. If the angle of projection was $36°$, find the initial speed of the projectile in m/s correct to the nearest integer.

12 A ball was projected from ground level and, when at its highest point, it just cleared a 3-metre wall. Given that the initial velocity of the ball was 20 m/s at an angle of α to the horizontal, prove that $\alpha = \sin^{-1}\dfrac{\sqrt{15}}{10}$.

13 Two particles P_1 and P_2 are projected from the origin with initial velocity vectors $20\underset{\sim}{i} + 30\underset{\sim}{j}$ and $60\underset{\sim}{i} + 50\underset{\sim}{j}$ respectively. If P_2 is projected 2 seconds after P_1, determine whether the particles collide and, if so, when they collide.

14 A particle is projected from the origin with initial speed $20\sqrt{2}$ m/s. On its flight it passes through the point $(20, 15)$ at time t. Let θ be the angle of projection.

a Show that $\sqrt{2}\,t\cos\theta = 1$ and $4\sqrt{2}\,t\sin\theta - t^2 = 3$.

b Hence show that $\tan^2\theta - 8\tan\theta + 7 = 0$.

c Hence find, correct to the nearest minute where necessary, the two possible values of θ.

15 A particle is projected with initial speed 50 m/s. On its flight it passes through a point P that is at a horizontal distance of 100 m and a vertical distance of 25 m from the point of projection.

a Show that the angle of projection α is $\tan^{-1}\frac{1}{2}$ or $\tan^{-1}\frac{9}{2}$.

b For each of the possible angles of projection, find: (i) the time it takes for the particle to reach P, (ii) the velocity of the particle as it passes through P, giving answers correct to one decimal place where necessary.

16 A particle is projected from level ground with initial speed V m/s at an angle of θ to the horizontal.

a Prove that: (i) the greatest height is $\dfrac{V^2 \sin^2 \theta}{2g}$ metres, (ii) the horizontal range is $\dfrac{V^2 \sin 2\theta}{g}$ metres.

b If the horizontal range is five times the greatest height, prove that $\theta = \arctan \frac{4}{5}$.

ENRICHMENT

17 A particle is projected from the floor of a horizontal tunnel that is 2 m high. If the particle just touches the ceiling of the tunnel, prove that the horizontal range inside the tunnel is $\sqrt{\dfrac{16}{g}(V^2 - 4g)}$ metres.

18 A projectile is fired from level ground with initial speed V at an angle of θ to the horizontal. Suppose that the greatest height of the projectile is h.

a Prove that $V^2 \sin^2 \theta = 2gh$.

b Prove that the particle is at height $\frac{h}{2}$ when $t = \dfrac{\left(\sqrt{2}+1\right)\sqrt{h}}{\sqrt{g}}$ or $t = \dfrac{\left(\sqrt{2}-1\right)\sqrt{h}}{\sqrt{g}}$.

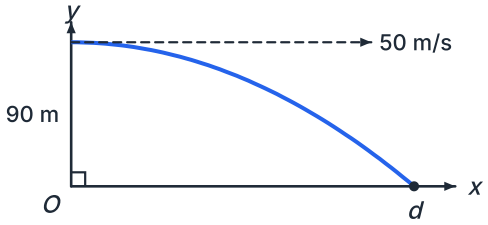
c If the ratio of the speed of the particle at height $\frac{h}{2}$ to the speed at height h is $\sqrt{5} : \sqrt{2}$, find θ.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q7** **1 mark**

The diagram below shows the trajectory of a ball thrown horizontally, at a speed of 50 ms^{-1}, from the top of a tower 90 metres above ground level.



After what time does the ball strike the ground?

(A) $3\sqrt{\frac{5}{g}}$ seconds

(B) $3\sqrt{\frac{50}{g}}$ seconds

(C) $6\sqrt{\frac{5}{g}}$ seconds

(D) $6\sqrt{\frac{50}{g}}$ seconds

Hint: thrown horizontally means $\dot{y} = 0$ when $t = 0$. Start from $\ddot{y} = -g$ and find when $y = 0$.

PP 2 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(e)** **4 marks**

A projectile is fired from horizontal ground such that horizontal and vertical displacements (in metres) from the point of projection, at time t seconds, are given by $x = \left(30\sqrt{3}\right)t$ and $y = 30t - 5t^2$.

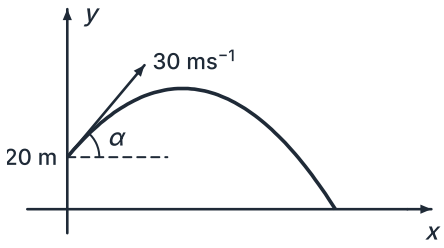
(i) Find the angle at which the projectile was fired. 3

(ii) Find the speed at which the projectile was fired. 1

Hint: differentiate to find $\dot{x}$ and $\dot{y}$, then put $t = 0$. Draw the velocity triangle.

PP 3 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)(i)(ii)** 4 marks

A projectile is fired from a point 20 metres above level ground, with a velocity of 30 ms^{-1} and at an angle of α° above the horizontal.



The displacement, velocity and acceleration of the projectile at any time are given by the displacement vector $\underset{\sim}{r}$, the velocity vector $\underset{\sim}{v}$ and the acceleration vector $\underset{\sim}{a}$ respectively. The initial velocity of the projectile is represented by the vector $\underset{\sim}{u}$.

The acceleration vector is $\underset{\sim}{a} = -10\underset{\sim}{j}$ and has units of ms^{-2}.

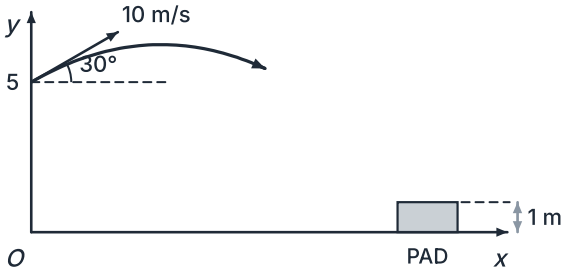
(i) Write down, in component form, the initial velocity vector $\underset{\sim}{u}$. 1

(ii) Derive the displacement vector $\underset{\sim}{r}$ in component form. 3

Hint: in (i), resolve the 30 ms^{-1} velocity. In (ii), integrate $\underset{\sim}{a}$ twice. Use $\underset{\sim}{u}$, and the starting position $20\underset{\sim}{j}$, to find the constants.

PP 4 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q12(d)** 6 marks

From a point five metres above level ground a stuntman is projected at an angle of elevation of 30° and at an initial speed of 10 m/s. A one metre high pad is placed on the ground to cushion his impact.



The displacement vector function is

$$\underset{\sim}{r} = 5\sqrt{3}t\,\underset{\sim}{i} + 5\left(1 + t - t^2\right)\underset{\sim}{j}. \qquad \text{(DO NOT SHOW THIS)}$$

(i) Show that the time of flight for the stuntman to land on the pad is approximately 1.52 seconds. 1

(ii) Determine the distance from O, correct to the nearest centimetre, where the pad should be centred, to hopefully cushion his impact. 1

(iii) Calculate his impact speed, to the nearest m/s. 2

(iv) This is to be performed at an indoor stadium with a ceiling height of 8 metres. Calculate the distance the stuntman will miss the ceiling by at his highest point. 2

Hint: find where the stuntman lands (on the pad). Differentiate $\underset{\sim}{r}$ to get $\dot{x}$ and $\dot{y}$. At the highest point, $\dot{y} = 0$.

PP 5 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(b)** 5 marks

A firework is projected from the point $A(0, 10)$ with initial velocity of magnitude V m s^{-1} at an angle θ to the horizontal.

Before reaching its maximum height, the firework explodes at a point B.

The time taken to reach B was 2 seconds.

Its velocity at B is 34 ms^{-1} where its path makes an angle α with the horizontal such that $\tan \alpha = \dfrac{8}{15}$.

The vector position of the firework is given by $\underset{\sim}{r} = (Vt\cos\theta)\underset{\sim}{i} + \left(10 + Vt\sin\theta - 5t^2\right)\underset{\sim}{j}$.

(i) Show that $V\sin\theta = 36$. 3

(ii) Find V, the magnitude of the initial velocity, and the size of the angle of projection θ. 2

Hint: resolve the velocity at B into its components $34\cos\alpha$ and $34\sin\alpha$. Compare them with $\dot{x}$ and $\dot{y}$ at $t = 2$.

LESSON 10B

Projectile motion: the equation of path

Eliminating t from the time equations gives the Cartesian equation of the path, which is always an upside-down parabola. This lesson covers deriving the equation of path, using it to find heights, range and direction, the range formula $\frac{V^2}{g}\sin 2\alpha$, and the equation of path as a quadratic in $\tan\alpha$.

CONCEPT 1 · deriving the equation of path

THE GENERAL CASE

The formulae for x and y in terms of t are a parametric equation of the path. A projectile fired from the origin with initial speed V and angle of elevation α has $\dot{x} = V\cos\alpha$, $\dot{y} = V\sin\alpha$ initially, and integrating $\ddot{x} = 0$ and $\ddot{y} = -g$ gives

$$x = Vt\cos\alpha \quad (3) \qquad\qquad y = -\tfrac{1}{2}gt^2 + Vt\sin\alpha \quad (6)$$

From (3), $t = \dfrac{x}{V\cos\alpha}$. Substituting into (6),

$$y = -\frac{gx^2}{2V^2\cos^2\alpha} + \frac{Vx\sin\alpha}{V\cos\alpha} \qquad \text{so} \qquad y = -\frac{gx^2}{2V^2}\sec^2\alpha + x\tan\alpha.$$

The Pythagorean identity $\dfrac{1}{\cos^2\alpha} = \sec^2\alpha = 1 + \tan^2\alpha$ gives the alternative form

$$y = -\frac{gx^2}{2V^2}\left(1 + \tan^2\alpha\right) + x\tan\alpha.$$

- This formula is **not to be memorised**: the working must always be shown, unless the equation is given in the question.
- The last form is quadratic in x, quadratic in $\tan\alpha$, and linear in g and y.
- Differentiating the equation of path gives the gradient $\dfrac{dy}{dx}$ at any value of x. This is an alternative way to find the angle of inclination in flight: $\tan\theta = \dfrac{dy}{dx}$.

TUTOR DEMO **From parametric to Cartesian**

A particle is projected from the origin, and $x = 30t$ and $y = -5t^2 + 40t$.

a Find the Cartesian equation of the path.

b Use it to find the horizontal range and the greatest height.

c Find $\dfrac{dy}{dx}$, and hence the angle of projection, to the nearest minute.

YOUR TURN **Same method**

A particle is projected from the origin, and $x = 20t$ and $y = -5t^2 + 15t$. Find the Cartesian equation of its path, the horizontal range, the greatest height, and the angle of projection to the nearest minute.

WORKED EXAMPLE 1 **Keep for homework**

A ball is thrown from the origin with speed 20 m/s at $60°$ to the horizontal. Take $g = 10$ m/s².

a Derive the Cartesian equation of the path.

b Find the horizontal range.

c Find the direction of motion when $x = 15\sqrt{3}$.

SOLUTION

a Initially $\dot{x} = 20\cos 60° = 10$ and $\dot{y} = 20\sin 60° = 10\sqrt{3}$.

From $\ddot{x} = 0$: $\dot{x} = 10$, and $x = 10t$ (using $x = 0$ when $t = 0$).

From $\ddot{y} = -10$: $\dot{y} = -10t + 10\sqrt{3}$, and $y = -5t^2 + 10\sqrt{3}\,t$.

From $x = 10t$, $t = \dfrac{x}{10}$. Substituting,

$$y = -5 \times \frac{x^2}{100} + 10\sqrt{3} \times \frac{x}{10} = -\frac{x^2}{20} + \sqrt{3}\,x.$$

b Put $y = 0$: $x\left(\sqrt{3} - \dfrac{x}{20}\right) = 0$, so $x = 0$ or $x = 20\sqrt{3}$. The range is $20\sqrt{3} \doteq 34.6$ m.

c Differentiating,

$$\frac{dy}{dx} = -\frac{x}{10} + \sqrt{3}.$$

When $x = 15\sqrt{3}$,

$$\frac{dy}{dx} = -\tfrac{3}{2}\sqrt{3} + \sqrt{3} = -\frac{\sqrt{3}}{2}.$$

So $\tan\theta = -\dfrac{\sqrt{3}}{2}$ and $\theta \doteq -40°54'$. The ball is moving at about $40°54'$ below the horizontal.

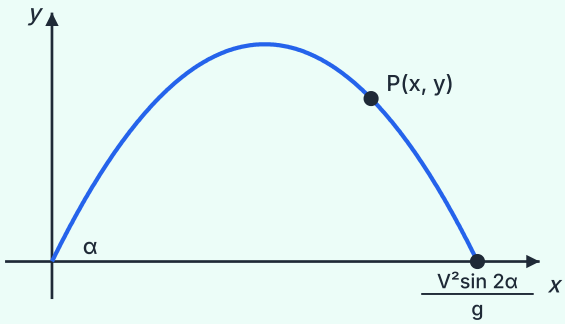
CONCEPT 2 · the range on level ground, and the maximum range

RANGE AND GREATEST HEIGHT

Putting $y = 0$ in the equation of path, $x = 0$ or

$$\frac{gx}{2V^2\cos^2\alpha} = \frac{\sin\alpha}{\cos\alpha}, \quad \text{so} \quad x = \frac{2V^2}{g}\sin\alpha\cos\alpha = \frac{V^2}{g}\sin 2\alpha.$$

- The range on level ground is $\frac{V^2}{g}\sin 2\alpha$.
- Because $\sin 2\alpha$ has a maximum value of 1 when $\alpha = 45°$, the **maximum range** is $\frac{V^2}{g}$, when $\alpha = 45°$.
- Any range less than the maximum can be reached with two angles, α and $90° - \alpha$, because $\sin 2\alpha = \sin(180° - 2\alpha)$.
- By symmetry, the greatest height is reached halfway, at $x = \frac{V^2\sin 2\alpha}{2g}$, and it is $\frac{V^2\sin^2\alpha}{2g}$.



These results must be derived in an exam unless they are given.

TUTOR DEMO Using the range

a Using the equation of path, show that the range on level ground is $\frac{V^2}{g}\sin 2\alpha$.

b A ball is kicked from level ground at 30 m/s and lands 45 m away. Taking $g = 10$ m/s², find the two possible angles of projection, and the maximum possible range at this speed.

YOUR TURN **Same method**

A stone is thrown from level ground at 20 m/s and lands $20\sqrt{3}$ m away. Taking $g = 10$ m/s² and assuming the range formula, find the two possible angles of projection, and the greatest height for each.

WORKED EXAMPLE 2 **Keep for homework**

A particle is projected from the origin with initial speed V at an angle θ to the horizontal. You may assume that its path is $y = x\tan\theta - \dfrac{gx^2\sec^2\theta}{2V^2}$. Find $\dfrac{dy}{dx}$, and hence show that the vertex of the parabola is $\left(\dfrac{V^2\sin 2\theta}{2g}, \dfrac{V^2\sin^2\theta}{2g}\right)$.

SOLUTION

$$\frac{dy}{dx} = \tan\theta - \frac{gx\sec^2\theta}{V^2}$$

At the vertex the path is horizontal, so put $\dfrac{dy}{dx} = 0$:

$$x = \frac{V^2\tan\theta}{g\sec^2\theta} = \frac{V^2}{g}\sin\theta\cos\theta = \frac{V^2\sin 2\theta}{2g}.$$

Substituting $x = \dfrac{V^2}{g}\sin\theta\cos\theta$ into the two terms of the equation of path:

$$x\tan\theta = \frac{V^2\sin^2\theta}{g} \qquad \text{and}$$

$$\frac{gx^2\sec^2\theta}{2V^2} = \frac{g}{2V^2}\times\frac{V^4\sin^2\theta\cos^2\theta}{g^2\cos^2\theta} = \frac{V^2\sin^2\theta}{2g},$$

so $y = \dfrac{V^2\sin^2\theta}{g} - \dfrac{V^2\sin^2\theta}{2g} = \dfrac{V^2\sin^2\theta}{2g}$, as required.

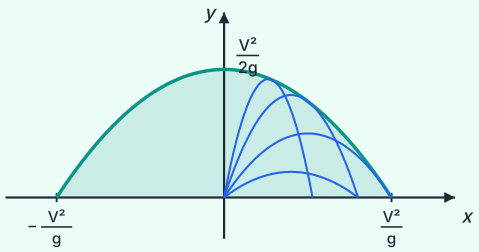
CONCEPT 3 · the equation of path as a quadratic in tan α

WHICH POINTS CAN BE HIT?

Multiplying the equation of path through by $2V^2$ and rearranging gives a quadratic in $\tan\alpha$:

$$gx^2\tan^2\alpha - 2V^2x\tan\alpha + \left(2V^2y + gx^2\right) = 0.$$

- To hit a given point $P(x, y)$ at a given speed, substitute x and y and solve for $\tan\alpha$. Two roots give the low shot and the lobbed shot.
- There is a solution if and only if $\Delta \geq 0$, which simplifies to $2V^2gy \leq V^4 - g^2x^2$.
- Geometrically, P can be reached only if it lies on or under the parabola $y = \dfrac{V^4 - g^2x^2}{2V^2g}$, whose vertex $\left(0, \dfrac{V^2}{2g}\right)$ lies on the y-axis and whose zeroes are $x = \pm\dfrac{V^2}{g}$.



Paths at one speed and several angles. None goes above the outer (shaded) parabola.

TUTOR DEMO — Hitting a point at a fixed speed

A projectile is fired from the origin at 20 m/s at an angle α. Take $g = 10$ m/s².

a Show that its path is $y = x\tan\alpha - \dfrac{x^2}{80}\left(1 + \tan^2\alpha\right)$.

b Find the two angles, to the nearest degree, at which it can be fired to pass through $(20, 5)$.

c Show that it cannot pass through $(40, 10)$ for any angle.

YOUR TURN **Same method**

A projectile is fired from the origin at 30 m/s at an angle α. Take $g = 10$ m/s² and assume the path is $y = x\tan\alpha - \dfrac{x^2}{180}\left(1+\tan^2\alpha\right)$. Find the two angles, to the nearest degree, at which it can pass through $(30, 10)$.

WORKED EXAMPLE 3 **Keep for homework**

Use the equation of path $y = -\dfrac{gx^2}{2V^2}\left(1+\tan^2\alpha\right) + x\tan\alpha$. Arrange it as a quadratic in $\tan\alpha$, and hence show that with a given initial speed V and variable angle α of elevation, a projectile can be fired through the point $P(x, y)$ if and only if $2V^2gy \le V^4 - g^2x^2$. What does this mean geometrically?

SOLUTION

Multiplying through by $2V^2$,

$$2V^2y = -gx^2\left(1+\tan^2\alpha\right) + 2V^2x\tan\alpha$$

$$gx^2\tan^2\alpha - 2V^2x\tan\alpha + \left(2V^2y + gx^2\right) = 0.$$

This is a quadratic in $\tan\alpha$, so it has a solution when $\Delta \ge 0$:

$$\left(2V^2x\right)^2 - 4gx^2\left(2V^2y + gx^2\right) \ge 0$$

$$4V^4x^2 - 8x^2V^2gy - 4g^2x^4 \ge 0$$

Dividing by $4x^2$: $2V^2gy \le V^4 - g^2x^2$.

Rearranging again,

$$y \le \frac{V^4 - g^2x^2}{2V^2g}.$$

This means that the target $P(x, y)$ must lie on or under the parabola $y = \dfrac{V^4 - g^2x^2}{2V^2g}$, whose vertex $\left(0, \dfrac{V^2}{2g}\right)$ lies on the y-axis, and whose zeroes are $x = \dfrac{V^2}{g}$ and $x = -\dfrac{V^2}{g}$.

Check: the zeroes agree with the maximum range $\dfrac{V^2}{g}$, and the vertex agrees with the greatest height of a vertical throw, $\dfrac{V^2}{2g}$.

TEXTBOOK HOMEWORK · Exercise 10B

Note: in this exercise take $g = 10$ m/s² unless otherwise indicated.

FOUNDATION

1 Suppose that a particle is projected from the origin and its parabolic path has equation $y = -\frac{5}{324}x^2 + \frac{4}{3}x$.

a Find the height of the particle when it has travelled a horizontal distance of 12 m.

b Find how far the particle has travelled horizontally when its height is 19 m.

c Find $\dfrac{dy}{dx}$ and hence show that: (i) the angle of projection is $\tan^{-1}\frac{4}{3}$, (ii) when the horizontal distance travelled is 18 m, the direction of motion is $\tan^{-1}\frac{7}{9}$ above the horizontal, (iii) when the horizontal distance travelled is 54 m, the direction of motion is $\tan^{-1}\frac{1}{3}$ below the horizontal.

2 A particle is projected from a point O. Its parabolic motion is governed by the parametric equations $x = 48t$ and $y = -5t^2 + 20t$, where x metres and y metres are the respective horizontal and vertical components of the displacement from O after t seconds.

a Show that the parabolic path has Cartesian equation $y = -\frac{5}{2304}x^2 + \frac{5}{12}x$.

b Use the Cartesian equation of the parabola to find: (i) the horizontal range of the particle, (ii) the greatest height of the particle, (iii) the angle of projection (correct to one decimal place), (iv) the direction in which the particle is moving when $x = 120$ (correct to the nearest degree).

3 A particle is projected from the origin O with initial velocity vector $3\underset{\sim}{i} + \underset{\sim}{j}$.

a What is the initial speed of the particle?

b Show that the parabolic path has parametric equations $x = 3t$ and $y = t - 5t^2$.

c Hence show that the Cartesian equation of the path is $9y = 3x - 5x^2$.

d Find, correct to one decimal place, the direction of motion when: (i) $x = 0.15$, (ii) $t = 0.15$.

4 An object is tossed from the top of a 20 m tower with initial velocity vector $5\underset{\sim}{i}$.

a If the point of projection is the origin, find the Cartesian equation of the path of the object.

b Find how far the object lands from the base of the tower.

c Find, correct to the nearest degree, the direction in which the object is travelling when it hits the ground.

DEVELOPMENT

5 A ball is tossed from the origin O that is 6 m above ground level. The initial velocity is 24 m/s at an angle of $30°$ above the horizontal.

a Find $\dot{x}$, $\dot{y}$, x and y as functions of t.

b Show that the Cartesian equation of the path of the ball is $y = \frac{1}{\sqrt{3}}x - \frac{5}{432}x^2$.

c If D metres is the horizontal distance that the ball has travelled when it strikes the ground, show that $5D^2 - 144\sqrt{3}\,D - 2592 = 0$.

d Hence find D correct to one decimal place.

6 A stone is thrown from the top of a 60 m cliff at an angle of $27°$ below the horizontal. It lands in the ocean 35 m from the base of the cliff.

a Taking the origin at the point of projection, show that the path of the stone has Cartesian equation $y = -x\tan 27° - \dfrac{5x^2}{V^2\cos^2 27°}$, where V is the initial speed.

b Hence find: (i) the initial speed of the stone correct to one decimal place, (ii) the direction in which the stone is moving, correct to the nearest degree, when it lands in the ocean.

7 A particle is projected from the origin with initial speed V at an angle of θ to the horizontal. You may assume that the equation of its path is $y = x\tan\theta - \dfrac{gx^2\sec^2\theta}{2V^2}$.

a Show that the horizontal range of the particle is $\dfrac{V^2\sin 2\theta}{g}$.

b If the initial speed is 30 m/s and the horizontal range is to be 75 m, find the two possible values of θ correct to one decimal place.

8 A particle is projected from the origin with initial speed V at an angle of θ to the horizontal. You may assume the equation of its parabolic path (as given, for example, in Question 7).

a Find $\dfrac{dy}{dx}$, and hence show that the vertex of the parabola is $\left(\dfrac{V^2\sin 2\theta}{2g}, \dfrac{V^2\sin^2\theta}{2g}\right)$.

b If the initial speed is 20 m/s and the greatest height is 15 m, find θ.

9 A gun can fire a shell with a constant initial speed V and a variable angle of elevation α. Assume that t seconds after being fired, the horizontal and vertical displacements x and y of the shell from the gun are given by $x = Vt\cos\alpha$ and $y = -\frac{1}{2}gt^2 + Vt\sin\alpha$.

a Show that the Cartesian equation of the shell's path may be written as $gx^2\tan^2\alpha - 2xV^2\tan\alpha + \left(2yV^2 + gx^2\right) = 0$.

b Suppose that $V = 200$ m/s, and that the shell hits a target positioned 3 km horizontally and 0.5 km vertically from the gun. Show that $\tan\alpha = \dfrac{4+\sqrt{3}}{3}$ or $\tan\alpha = \dfrac{4-\sqrt{3}}{3}$, and hence find the two possible values of α, correct to the nearest minute.

10 A particle is projected from a point O with speed 34 m/s at an angle of θ above the horizontal.

a Show that the parabolic path of the particle has equation $y = x\tan\theta - \dfrac{5x^2\sec^2\theta}{1156}$.

b During its flight the particle passes through a point that is 11 m above O and 30 m horizontally from O. Show that the two possible angles of projection are $\theta = \arctan\frac{8}{15}$ and $\theta = \arctan\frac{538}{75}$.

ENRICHMENT

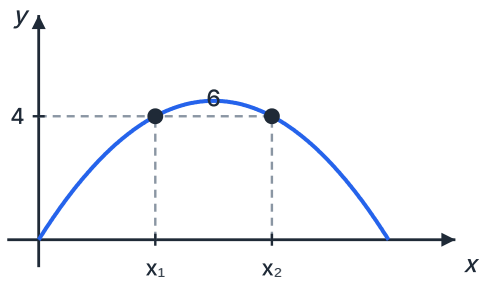
11 A boy throws a ball with speed V m/s at an angle of $45°$ to the horizontal.

a Derive expressions for the horizontal and vertical components of the displacement of the ball from the point of projection.

b Hence show that the Cartesian equation of the path of the ball is $y = x - \dfrac{gx^2}{V^2}$.

c The boy is now standing on a hill inclined at an angle θ to the horizontal. He throws the ball at the same angle of elevation of $45°$ and at the same speed of V m/s. If he can throw the ball 60 metres measured down the hill, but only 30 metres measured up the hill, use the result in part b to show that $\tan\theta = 1 - \dfrac{30g\cos\theta}{V^2} = \dfrac{60g\cos\theta}{V^2} - 1$, and hence that $\theta = \tan^{-1}\frac{1}{3}$.

12 A particle is projected from the origin with speed V m/s at an angle α to the horizontal.



a Assuming that the coordinates of the particle at time t are $\left(Vt\cos\alpha,\ Vt\sin\alpha - \frac{1}{2}gt^2\right)$, prove that the horizontal range R of the particle is $\dfrac{V^2\sin 2\alpha}{g}$.

b Hence prove that the path of the particle has equation $y = x\left(1 - \dfrac{x}{R}\right)\tan\alpha$.

c Suppose that $\alpha = 45^\circ$ and that the particle passes through two points 6 metres apart and 4 metres above the point of projection, as shown in the diagram. Let x_1 and x_2 be the x-coordinates of the two points. (i) Show that x_1 and x_2 are the roots of the equation $x^2 - Rx + 4R = 0$. (ii) Use the identity $(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_2x_1$ to find R.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q7** **1 mark**

A projectile has the equation of path $y = -3x^2 + 2x + 4$. How far will it have travelled horizontally before it returns to its original height?

(A) $\frac{1}{3}$ unit

(B) $\frac{2}{3}$ unit

(C) 2 units

(D) 4 units

Hint: the projectile starts where $x = 0$. Find its height there first.

PP 2 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q13(b)** **2 marks**

A particle is projected from a point O with velocity V m/s at an angle θ to the horizontal. At any time t seconds the horizontal and vertical components of displacement are given by $x = Vt\cos\theta$ and $y = Vt\sin\theta - \frac{1}{2}gt^2$ where g is the acceleration due to gravity.

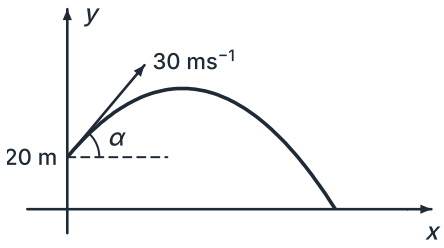
Show that the cartesian equation of the path is given by $y = x\tan\theta - \frac{gx^2}{2V^2}\left(1 + \tan^2\theta\right)$.

Hint: eliminate t. Which of the two equations is easy to solve for t?

PP 3 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)(iii)** 3 marks

A projectile is fired from a point 20 metres above level ground, with a velocity of 30 ms^{-1} and at an angle of α° above the horizontal.

The acceleration vector is $\underset{\sim}{a} = -10\underset{\sim}{j}$ and has units of ms^{-2}.



Earlier parts of this question (not included here) showed that the displacement vector is

$$\underset{\sim}{r} = 30t\cos\alpha^\circ\,\underset{\sim}{i} + \left(30t\sin\alpha^\circ - 5t^2 + 20\right)\underset{\sim}{j}.$$

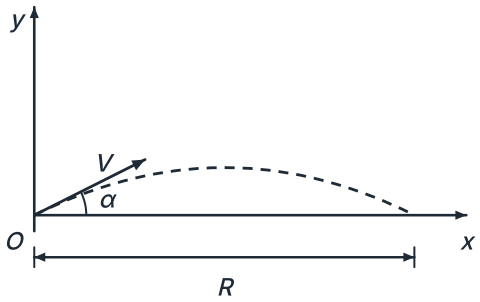
(iii) By showing that the Cartesian equation of the trajectory is $y = x\tan\alpha - \dfrac{x^2}{180}\sec^2\alpha + 20$, or otherwise, find the value(s) of α for which the projectile will hit a target 90 metres down-range and 1 metre above the ground. 3

Hint: make t the subject of the x equation. At the target, write $\sec^2\alpha$ in terms of $\tan\alpha$.

PP 4 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(d)** 6 marks

The displacement of a projectile fired from the origin O has a horizontal component x and a vertical component y.
The equations of motion are

$$x = Vt\cos\alpha, \quad y = Vt\sin\alpha - 5t^2,$$



where V is the initial speed in metres per second, α is the angle of projection as in the diagram, and t is the time in seconds. The range, R, is the total horizontal distance travelled by the projectile.

(i) Derive an expression for R in terms of V and α. 2

(ii) The maximum range for a given value of V is $R_{\max}$. Show that $R_{\max}$ occurs when $\alpha = 45°$ and obtain a fully simplified expression for it. 3

(iii) Show that if the projectile is to hit a target 12 km away, it must be launched at a speed faster than the speed of sound in air, which is 343 m/s. 1

Hint: the projectile lands when $y = 0$ and $t \neq 0$. In (iii), compare 12 km with $R_{\max}$.

PP 5 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(c)** 6 marks

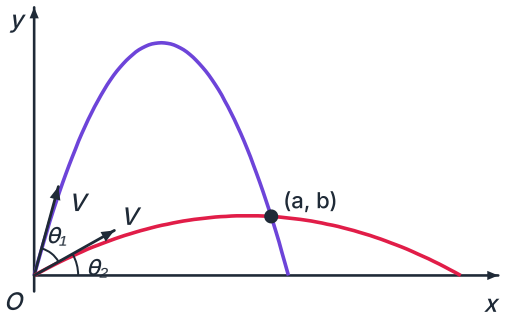
A ball is thrown from the origin, O, with a velocity, V m/s, at an angle of elevation of θ, where $\theta \neq 90°$. You may assume that the position vector $\underset{\sim}{r}(t)$ of the ball in metres, from O, after t seconds is given by:

$$\underset{\sim}{r}(t) = Vt\cos\theta\,\underset{\sim}{i} + \left(Vt\sin\theta - \frac{1}{2}gt^2\right)\underset{\sim}{j} \qquad \text{(Do not prove this)}$$

where $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are the unit vectors in the horizontal and vertical directions respectively.

(i) Show that the equation of flight of the ball is: $y = x\tan\theta - \dfrac{x^2}{4h}\left(1+\tan^2\theta\right)$, where $h = \dfrac{V^2}{2g}$. 2

(ii) Two balls are thrown simultaneously with the same velocity, V m/s, at angles of elevation of θ_1 and θ_2 as shown below. 2



Show that in order for the two balls to pass through the point (a, b): $a^2 < 4h(h-b)$.

(iii) Show that if two balls are thrown simultaneously with the same velocity, V m/s, and at angles of elevation of θ_1 and θ_2, such that $\theta_1 < 45°$ and $\theta_2 < 45°$, then the two balls will **not** pass through the point (a, b). 2

Hint: put (a, b) into the equation of flight and read it as a quadratic in $\tan\theta$. Each angle gives one root.

CHAPTER 10 REVIEW

Chapter review exercise

Mixed practice from the whole chapter. Take $g = 10$ m/s² unless otherwise indicated. Question 11 is from the Chapter 9 review and practises related rates.

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1 A particle is projected from horizontal ground with a speed of 60 m/s at an angle of 40° above the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 60\cos 40^\circ$ and $\dot{y} = -10t + 60\sin 40^\circ$.

b Hence find x and y in terms of t.

c Find, correct to one decimal place: (i) when the particle returns to the ground, (ii) the horizontal distance travelled by the particle, (iii) the greatest height reached above the ground.

2 A rock is thrown from the top of a vertical cliff of height 40 metres. Its initial velocity is 30 m/s at an angle of 60° above the horizontal. Find, correct to two decimal places where necessary: **a** the greatest height reached by the rock above the point of projection, **b** the distance that the rock lands from the base of the cliff, **c** the speed at which the rock hits the ground.

3 A particle is projected from the origin O with a speed of 25 m/s at an angle of $\tan^{-1}\frac{4}{3}$ to the horizontal.

a Show that $\dot{x}=15$ and $\dot{y}=-10t+20$.

b Find x and y as functions of t.

c Find the distance of the particle from the point of projection after one second.

d Show that the velocity of the particle after one second is $5\sqrt{13}$ m/s at an angle of $\tan^{-1}\frac{2}{3}$ above the horizontal.

e If R is the horizontal range of the particle and H is the greatest height, show that $R=3H$.

4 An object is projected from level ground with initial speed 10 m/s at an angle of 45° to the horizontal.

a Find $\dot{x}$, x, $\dot{y}$ and y by integration from $\ddot{x}=0$ and $\ddot{y}=-10$. Then, by eliminating t, show that the equation of the parabolic path is $y=x-\frac{1}{10}x^2$.

b Use the equation of the path to find the horizontal range and the maximum height.

c Suppose first that the stone hits a wall 8 metres away. (i) Find how far up the wall the stone hits. (ii) Differentiate the equation of the path, and hence find the direction of the object when it hits the wall.

d Suppose now that the stone hits a ceiling 2.1 metres high. (i) Find the horizontal distance travelled before impact. (ii) Find the angle at which the stone hits the ceiling.

5 A particle is projected from the origin. Its initial velocity vector is $24\underset{\sim}{i} + 18\underset{\sim}{j}$.

a Express its velocity at time t in component vector form.

b Express its displacement at time t in component vector form.

c (i) Find the initial speed of the particle. (ii) Find the position vector of the particle after 4 seconds. (iii) Find the position vector of the particle when it reaches its greatest height.

6 A ball is thrown with initial speed V at an angle of α to the horizontal. Two seconds later it passes through the point with position vector $24\sqrt{5}\,\underset{\sim}{i} + 28\underset{\sim}{j}$.

a Find the initial velocity vector of the ball.

b Find the position vector of the ball after 3 seconds.

c Is the ball rising or falling after 3 seconds? Justify your answer.

7 A particle is projected from a point O on level ground with initial speed V m/s at an angle of θ to the horizontal. Its horizontal range is 108 m, and its flight time is 3 seconds.

a Show that $V = 39$ m/s and $\theta = \arctan \frac{5}{12}$.

b Find the greatest height reached by the particle.

8 Steve tosses an apple to Adam, who is sitting near him. Adam catches the apple at exactly the same height that Steve released it. Suppose that the initial speed of the apple is $V = 5$ m/s, and the initial angle α of elevation is given by $\tan \alpha = 2$.

a Use a velocity resolution diagram to find the initial values of $\dot{x}$ and $\dot{y}$.

b Find $\dot{x}$, x, $\dot{y}$ and y by integrating $\ddot{x} = 0$ and $\ddot{y} = -10$, taking the origin at Steve's hands.

c Show by substitution into y that the apple is in the air for less than 1 second.

d Find the greatest height above the point of release reached by the apple.

e Show that the flight time is $\frac{2}{5}\sqrt{5}$ seconds, and hence find the horizontal distance travelled by the apple.

f Find $\dot{x}$ and $\dot{y}$ at the time Adam catches the apple. Then use a diagram to resolve the velocity and show that the final speed equals the initial speed, and the final angle of inclination is the opposite of the initial angle of elevation.

g The path of the apple is a parabolic arc. By eliminating t from the equations for x and y, find its equation in Cartesian form.

9 A golf ball was hit from level ground and just cleared a 12 m tall tree that was 30 m from the point of projection. If the angle of projection was $26°$, find the initial speed of the ball in m/s correct to the nearest integer.

10 A ball was thrown uphill from the base of a hill inclined at 30° to the horizontal. The initial velocity was 15 m/s at 60° to the horizontal.

a Show that the parabolic path of the ball has parametric equations $x = \frac{15}{2}t$ and $y = -5t^2 + \frac{15\sqrt{3}}{2}t$.

b Hence find the Cartesian equation of the path.

c (i) Find how far up the hill the ball landed (measuring along the slope), (ii) Find the time of flight.

11 A 5-metre ladder is leaning against a wall, and the base is sliding away from the wall at 5 cm/s. Find the rate at which: **a** the height, **b** the angle of inclination, is changing when the foot is 1.4 m from the wall.